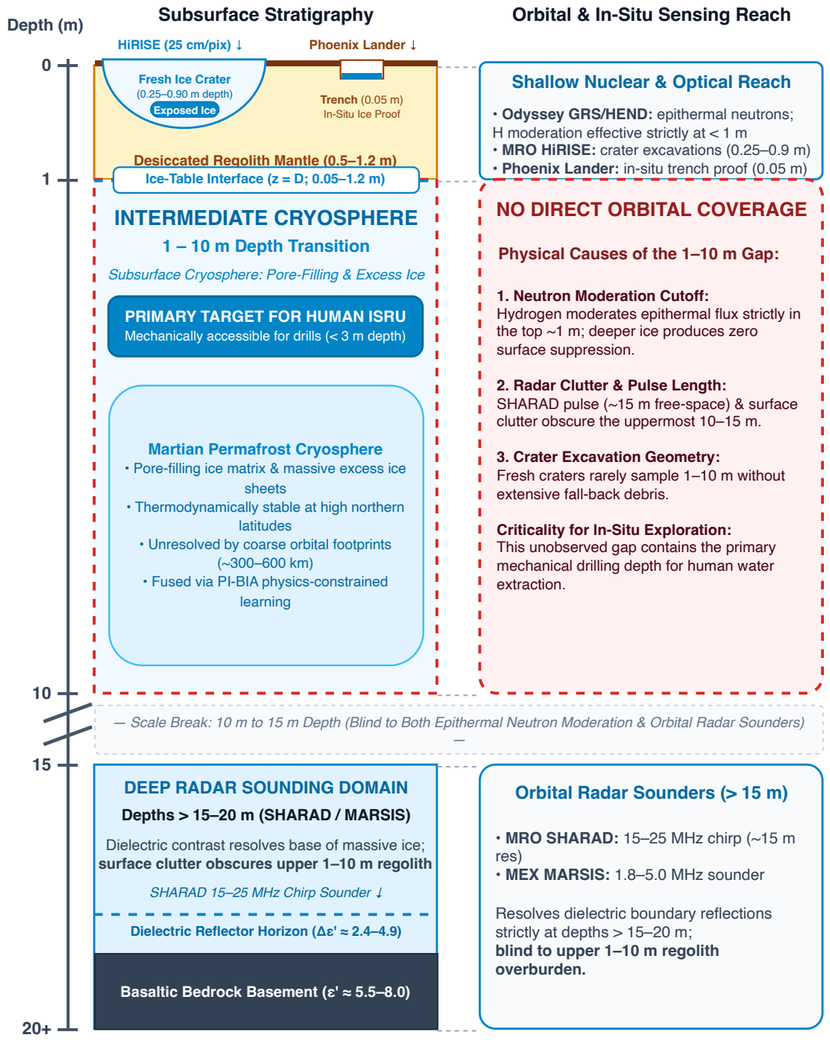
a Observation sensing regimes & spatial resolution scale

Geophysical sensing depth limits and multi-scale stratigraphy across the Martian northern plains



The PI-BIA Solution: Physics-constrained Bayesian downscaling bridges the multi-scale spatial gap by coupling epithermal neutron flux (< 1 m) with MOLA topography and empirical impact excavations.

b Physics-informed Bayesian inversion architecture (PI-BIA)

End-to-end framework integrating multi-modal planetary sensing, thermodynamic constraints, and uncertainty quantification

1 · Multi-Modal Inputs

NASA PDS Archives & In-Situ Benchmarks

Orbital Remote Sensing:

- **MGS MOLA MEGDR Topography**
  64 pix/deg global grid (926 m/pix)
  Elevation: –4995 to –2804 m
- **Mars Odyssey GRS / HEND**
  Corrected Epithermal Neutron Flux (CEPI)
  Spatial resolution: ~5° FWHM (~300 km)

Authentic Benchmarks (N = 26):

- **Fresh Ice Craters (n = 15)**
  HiRISE excavations (0.25–0.90 m)
- **Phoenix Lander Trench (n = 1)**
  In-situ ice anchor (0.05 m depth)
- **Non-Icy Controls (n = 10)**
  Verified dry craters (Dundas 2014 / HiRISE)

2 · Spatial Preprocessing

Co-Registered Multi-Modal Spatial Cube

Arcadia Study Corridor:

- Latitude: 37°N to 50°N | Lon: 180°E to 205°E
- High-res grid: 832 × 1600 cells (1.33M pixels)

Derived Geomorphology:

- **Elevation:** z from MOLA MEGDR datum
- **Slope gradient:** θ = arctan(|∇z|), mean: 0.36°
- **CEPI flux:** smooth bicubic to 64 pix/deg

Standardized Feature Vector:

- $\mathbf{x} = [\phi, \lambda, z, \theta, \Phi_{epi}]^T \in \mathbb{R}^5$ (ϕ: lat, Φ_{epi}: CEPI)
- Robust z-score: $\tilde{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$ (no feature clamping)

3 · PI-BIA Inversion

Dual-Stage Physics Engine & Uncertainty

Stage I: Ice Detection P(ice)

- Physics-Constrained Logistic Model
- Monotonicity Bounds: $\beta_{CEPI} \le 0$, $\beta_{lat} \ge 0$
- Bounded L-BFGS-B optimization (λ = 0.5)

Stage II: Depth Inversion D(x)

- Bayesian Ridge (calibrated on n = 16 ice sites)
- Predictive posterior: $D \sim \mathcal{N}(\mu_D, \sigma_D^2)$
- Physical lower bound: D ≥ 0.05 m

Laplace Epistemic Uncertainty:

- Laplace Hessian $\mathbf{H} = \nabla^2(-\log p)$
- Covariance: $\boldsymbol{\Sigma} = (\mathbf{H} + \lambda \mathbf{I})^{-1}$
- Epistemic variance: $\sigma_{prob}^2 = (\nabla P)^T \boldsymbol{\Sigma} (\nabla P)$

4 · Outputs & Validation

Regional Grids & Operational Artifacts

Regional Planetary Grids:

- Occurrence probability grid P(ice)
- Ice-table depth raster D(x, y) (m)
- Laplace epistemic uncertainty σ_{prob}
- Resource Accessibility Index (RAI): $RAI = P(ice) \cdot e^{-D/1.25} \cdot f(\theta) \cdot \eta_{EDL} \cdot (1 - 1.5\,\sigma_{prob})$

Candidate Landing Sites:

- Top-5 RAI-ranked landing sites
- Criteria: elevation < –3900 m, slope < 1.0°

Scientific Verification Deliverables:

- Complete data cube (**.npz** & **.json**)
- Stratified CV & LOOCV benchmark validation
- Performance metrics tabulated in scorecard below

Stratified 5-Fold & LOOCV Generalization Assessment

Mathematical Formulation & Bayesian Uncertainty Framework

$\beta_{CEPI} \le 0,\ \beta_{lat} \ge 0$ | $\mathbf{H} = \nabla^2(-\log p)$

1 · Physics-Constrained Logistic Ice Detection Classifier:

$$P(\text{ice} = 1 \mid \mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^T \tilde{\mathbf{x}}) = 1 / [1 + \exp(-(\beta_0 + \boldsymbol{\beta}^T \tilde{\mathbf{x}}))], \quad \text{s.t. } \beta_{CEPI} \le 0,\ \beta_{lat} \ge 0$$

$$\mathcal{L}(\boldsymbol{\beta}, \beta_0) = -\sum_{i=1}^{N} [\, y_i \ln P_i + (1 - y_i) \ln(1 - P_i)\,] + (\lambda/2)\, \|\boldsymbol{\beta}\|_2^2 \quad (\lambda = 0.5, \text{ bounded L-BFGS-B})$$

Standardized features $\tilde{\mathbf{x}} = [\text{lat}, \text{lon}, z_{MOLA}, \theta_{slope}, \Phi_{CEPI}]^T \in \mathbb{R}^5$ with physical gradient monotonicity

2 · Laplace Epistemic Uncertainty & Bayesian Depth Inversion:

$$\boldsymbol{\Sigma}_{Laplace} = (\mathbf{H} + \lambda \mathbf{I})^{-1}, \quad \mathbf{H} = \nabla_{\boldsymbol{\theta}}^2(-\ln p(\mathcal{D} \mid \boldsymbol{\theta}))|_{\hat{\boldsymbol{\theta}}_{MAP}}, \quad \sigma_{prob}^2(\mathbf{x}) = (\nabla_{\boldsymbol{\theta}} P)^T \boldsymbol{\Sigma}_{Laplace} (\nabla_{\boldsymbol{\theta}} P)$$

Stage II Analytical Posterior: $\boldsymbol{\Sigma}_w = (\lambda_{noise}\, \mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1}, \quad \boldsymbol{\mu}_w = \lambda_{noise}\, \boldsymbol{\Sigma}_w\, \mathbf{X}^T\, \mathbf{y}_{depth}$

$D(\mathbf{x}) \sim \mathcal{N}(\tilde{\mathbf{x}}^T \boldsymbol{\mu}_w,\ \sigma_{noise}^2 + \tilde{\mathbf{x}}^T \boldsymbol{\Sigma}_w \tilde{\mathbf{x}})$ **with physical permafrost bound** $D(\mathbf{x}) \ge 0.05$ m

3 · Operational Resource Accessibility Index (RAI):

$$RAI(\mathbf{x}) = P(\text{ice} = 1 \mid \mathbf{x}) \cdot \exp(-D(\mathbf{x}) / 1.25\ \text{m}) \cdot f_{slope}(\theta) \cdot \eta_{EDL}(z) \cdot (1 - 1.5\, \sigma_{prob}(\mathbf{x}))$$

where $f_{slope}(\theta) = \max(1 - \theta / 10°, 0)$, $\eta_{EDL}(z) = \text{clip}((-z - 1500\ \text{m}) / 2500\ \text{m}, 0, 1)$

$(1 - 1.5\, \sigma_{prob})$ penalizes epistemic variance; $\exp(-D / 1.25\ \text{m})$ models mechanical drilling limit ($D_0 = 1.25$ m)

Empirical Validation Scorecard

Calibrated on N = 26 authentic Martian benchmark sites

VALIDATION METRIC	VALUE	Method / Scale
Benchmark Sites	**N = 26**	16 ice / 10 dry
ROC-AUC	**0.931**	Binary discrimination
In-Sample F1	**0.867**	Decision thresh P = 0.50
5-Fold Stratified CV F1	**0.786 ± 0.072**	Out-of-fold generalization
Depth LOOCV RMSE	**0.0955 m**	Leave-one-out CV (9.6 cm)
Depth In-Sample MAE	**0.0449 m**	Mean absolute error (4.5 cm)
Calibration ECE	**0.123**	10-bin expected error
Laplace Epistemic σ	**0.146 ± 0.047**	Dynamic posterior

Top Landing Candidate: Arcadia Site #1
48.49°N, 190.03°E | Elev: –4071 m | Depth: 0.52 m | RAI: 0.533
Primary target: ultra-flat plain, max ice probability, sub-meter access